AP PRECALCULUS

Unit 1 • Worksheet 1.1-1.6D • Answer key

Review 1.1-1.6D • Review of Topics 1.1 to 1.6

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1 (Free response • AP style)

NO CALCULATOR

The polynomial function h has degree 3 and leading coefficient 1.

x	-2	-1	0	1	2	3
$h(x)$	-1	3	1	-1	3	19

(A) (i) Find the average rate of change of h over $[-1, 1]$.
(ii) How many real zeros does h have? Justify using the table.

(B) (i) Use the average rates of change over $[1, 2]$ and $[2, 3]$ to decide whether the graph of h is concave up or down on $[1, 3]$.
(ii) Write a limit statement for the end behavior of h as $x \to -\infty$.

(C) Let $k(x) = h(x - 2) + 4$. Find $k(3)$.

Solution

(A)(i) $\frac{-1-3}{2} = -2$. (ii) Sign changes on $[-2, -1]$, $[0, 1]$, $[1, 2]$; h is continuous, so at least 3 zeros; degree 3, so exactly 3.
(B)(i) 4 and 16: increasing rates, concave up. (ii) $\lim_{x \to -\infty} h(x) = -\infty$.
(C) $k(3) = h(1) + 4 = 3$.

Question 2 (Free response • AP style)

NO CALCULATOR

Let $p(x) = 2x^3 + x^2 - 13x + 6$.

(A) Show that $x = 2$ is a zero of p and write p as a product of linear factors.

(B) Describe the behavior of the graph at each zero and the end behavior of p.

(C) Find the average rate of change of p on $[-3, 2]$, and interpret its value using the graph.

Solution

(A) $p(2) = 16 + 4 - 26 + 6 = 0$. Division by $(x - 2)$ gives $2x^2 + 5x - 3 = (2x - 1)(x + 3)$. $p(x) = (x - 2)(2x - 1)(x + 3)$.
(B) Each zero (2, $\frac{1}{2}$, -3) has multiplicity 1: the graph crosses. $\lim_{x \to \infty} p = \infty$, $\lim_{x \to -\infty} p = -\infty$.
(C) $p(-3) = 0$ and $p(2) = 0$: the average rate of change is 0; the secant line through the two zeros is the x-axis.